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Applications of AI and ML: NLP in Everyday Life

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Today's Focus

A Simple Look At How AI Fits Into Our Daily World

1. **AI Around You**
   Spotting AI in everyday tasks

2. **NLP Applications in the Real World**
   How natural language tools help us day-to-day

3. **Live Demos and Clear Visuals**
   Watch how AI works with easy examples

4. **AI Limitations & Machine Learning**
   Understand its limits and how it learns

5. **Interactive Elements & Next Steps**
   Join quizzes, ask questions and see what's coming

Where Do You See Language AI?

Everyday Tools Powered By Language-based AI

AI is already integrated into many aspects of our digital lives:

- WhatsApp blocks suspicious messages
- Google guesses your search as you type
- Chatbots assist with queries and product suggestions
- YouTube adds real-time captions

What other examples have you noticed?

Can You Guess What AI's Doing?

Let's See How AI Understands Language

1	2
"Congratulations! You have won a free trip to Goa" Is it spotting spam? What clues is it using?	**"I need to book a cab to the airport"** Is it detecting intent or recognising a command?
3	**4**
"Elon Musk visited India in June" Can it find names, places or dates?	**ChatGPT is transforming how we write"** Is it checking tone, topic or tech terms?

NLP: How AI Understands Language

Techniques Used To Read And Respond To Human Language

Natural Language Processing helps computers read, understand and generate human language

- **Text Classification:** Spotting spam or emotion in text
- **Named Entity Recognition:** Finding names, places, and brands
- **Summarisation:** Picking out the key points from long text
- **Translation:** Switching text between languages
- **Question Answering:** Giving useful replies to questions

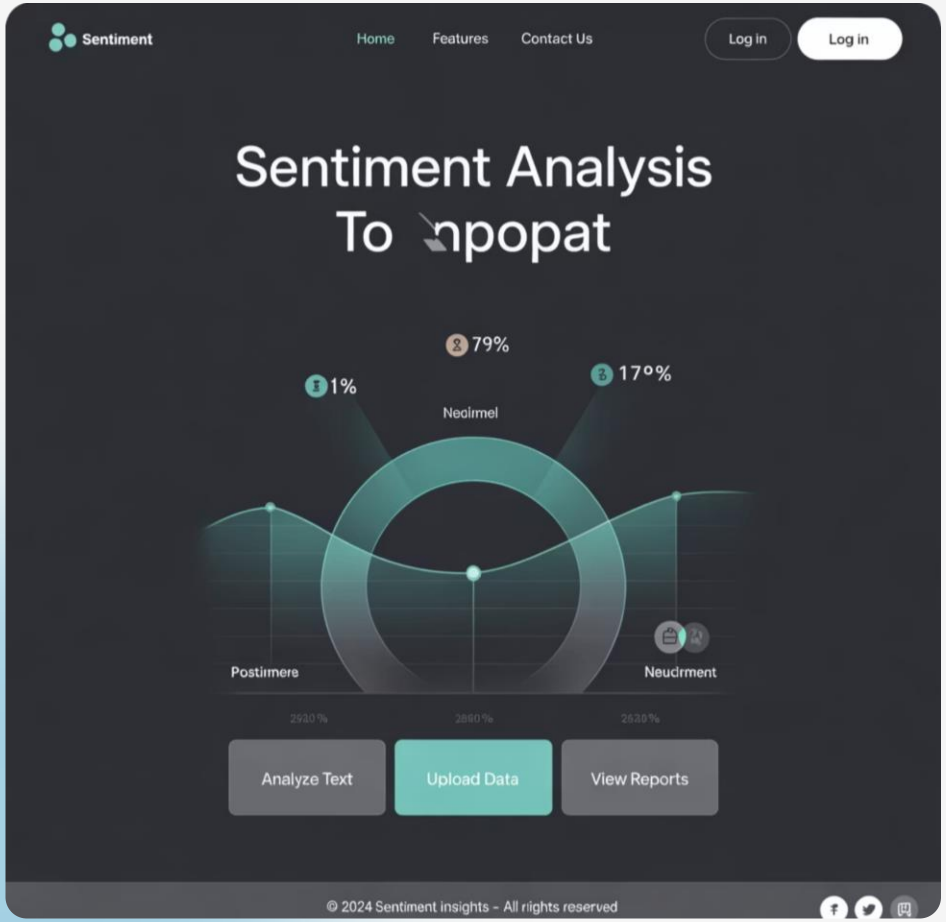
NLP in Everyday Use

Real-life Examples Of How AI Understands Language

Application	Real-World Example	How It Works
Spam Detection	Gmail filtering suspicious messages	Analyses text patterns typical of spam
Sentiment Analysis	Amazon product review ratings	Detects positive/negative language
Named Entity Recognition	News apps highlighting people/places	Identifies proper nouns and categories
Summarisation	News app's "TL;DR" feature	Extracts key information from text
Translation	Google Translate	Maps words/phrases between languages
Chatbots	Website customer support	Understands queries and generates responses

Sentiment Analysis in Action

AI Reading The Emotion In Reviews



"This phone is amazing, battery lasts all day!"
→ **AI prediction:** Positive *(92% sure)*

"The food was cold and service was slow."
→ **AI prediction:** Negative *(85% sure)*

The AI looks at words, tone, and context to judge the mood

Try it here!

Demo: Finding Names and Places

Example 1
"Sachin Tendulkar scored a century in Mumbai."
Person: Sachin Tendulkar | Location: Mumbai

Try it here!

Example 2
"Google is opening a new office in Hyderabad."
Organisation: Google | Location: Hyderabad

Try it here!

Automatic Text Summarisation

AI That Shortens Long Text While Keeping The Main Points

Original Text:

Apple today unveiled new MacBooks with advanced M4 chips, a thinner design, and improved battery life. The devices will be available starting next month. The company claims these are the fastest laptops they've ever produced, with significant improvements in performance for creative professionals. The new lineup includes both 14-inch and 16-inch models, with prices starting at $1,299."

Try it here!

Demo: Machine Translation

Smart Translations That Understand Meaning, Not Just Words

English → Spanish
Input: "How are you?"
Output: "¿Cómo estás?"

English → French
Input: "Good morning"
Output: "Bonjour"

English → Hindi
Input: "Thank you for your help"
Output: "आपकी मदद के लिए धन्यवाद"

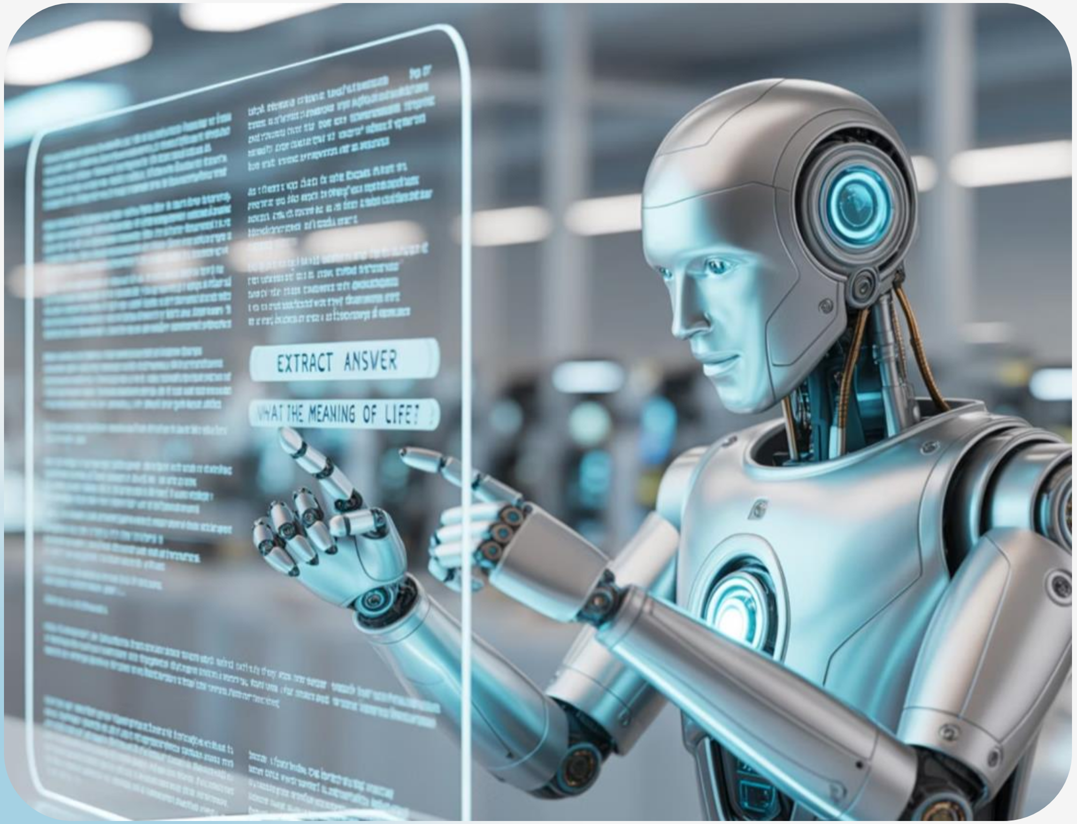
English → Japanese
Input: "I love learning about AI"
Output: "AIについて学ぶのが大好きです"

Translation AI understands context and idiomatic expressions, not just word-for-word conversion

Try it here!

Demo: Question Answering

Ask Questions. Get Answers. Instantly



How It Works:

AI reads text, finds key details and gives the right answer

Example

- **Context:** "The capital of France is Paris. It is known for the Eiffel Tower and the Louvre Museum."
- **Question**: "What is the capital of France?"
- **AI Output:** "Paris"

Upload *“Attention is All You Need”* PDF and ask questions about it

Try it here!

AI as a Black Box

A Simple View Of How AI Works: Input, Process, Output

Input
Raw text from users, documents or other data

Processing
AI detects patterns and applies learned rules

Output
Predictions, classifications or generated responses

AI uses past training to predict the most likely outcome

How to Imagine NLP Tasks

Simple Ways To Picture How NLP Works

Sentiment Analysis
Like giving a thumbs up or down based on tone

Named Entity Recognition
Like highlighting names and places

Summarisation
Like picking out and joining key points

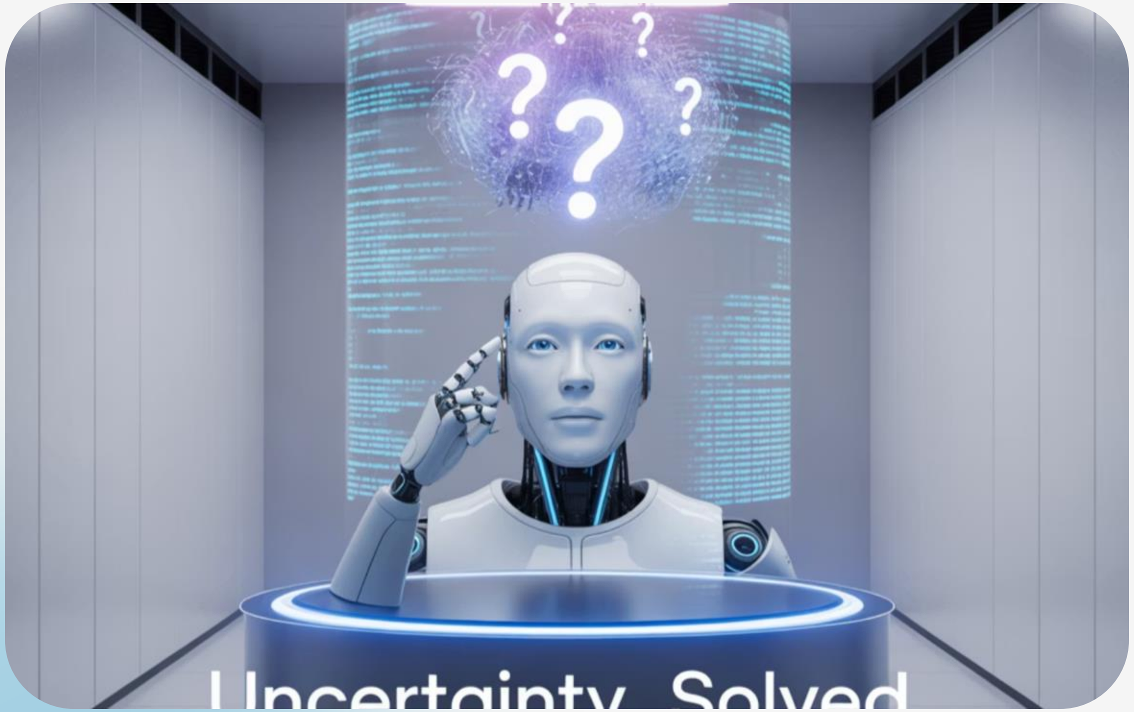
Translation
Like a translator switching between languages

Question Answering
Like using search to find exact answers

Text Classification
Like sorting mail into folders by topic

Funny and Tricky Mistakes

When AI Gets It Wrong:



- **Sarcasm:** "Oh great, another rainy day!" → AI reads it as **positive**(but it's actually negative)
- **Ambiguity:** "Apple is releasing new products." → Fruit or company?
- **Idioms:** "It’s raining cats and dogs." → Taken literally
- **Typos:** "I luv this phoen" → Confuses the AI
- **Cultural References:** "So bad it was good." → Mixed signals

Limitations of AI in NLP

Why AI Still Struggles With Human Communication

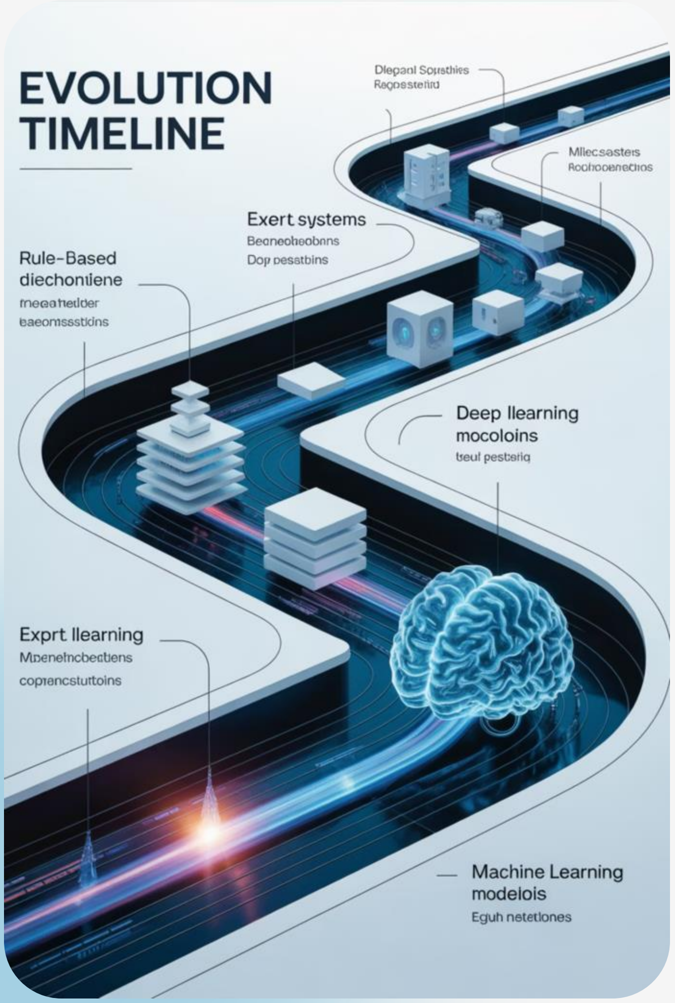
Language Complexity
Words can have many meanings and subtle context

Data Limitations
Biased or limited training data affects results

Context Understanding
Misses humour, sarcasm, and shared human experiences

From Rules to Learning Patterns

How Ai's Approach To Language Has Evolved Over Time



- **Rule-Based Era (1950s–1990s):**

  Programmers manually wrote explicit rules:

  - If word = "free" AND word = "win" → likely spam
  - If sentence ends with "?" → likely a question

  Limited by the number of rules humans could create

- **Statistical Era (1990s-2010s)**

  Systems began learning probability patterns from data:

  - Word frequencies and co-occurrences
  - Statistical models of language

  Better, but still limited in understanding context.

From Rules to Learning Patterns

How Ai's Approach To Language Has Evolved Over Time

Deep Learning Era (2010s-Present)

Models learn complex patterns from massive datasets:

Neural networks process language more like humans

Contextual understanding of words and phrases

Models with billions of parameters capture subtle nuances

What Did We Learn Today?

A Quick Recap Of Key Takeaways On NLP

Core NLP Tasks: Classification, entity recognition, summarisation, translation, Q&A

Tech Evolution: From rule-based to advanced machine learning

AI Limitations: Still struggles with sarcasm, context and culture

Rapid Progress: New models are improving fast

Everyday Use: NLP powers many tools we use daily

Questions and Curiosities

Let’s Reflect And Explore Together

- What surprised you most today?
- Which NLP tool do you use most?
- Any ethical concerns about AI in language?
- How might these tools change in 5–10 years?
- What language challenges will AI still face?

Add your questions to our **Curiosity Bank**!

Next Week: AI Beyond Text

Computer Vision

- Image classification and object detection
- Face recognition
- Understanding scenes

Video Processing

- Motion tracking
- Content moderation
- Smart video editing

Speech Technology

- Speech-to-text
- Voice assistants
- Speaker recognition

Discover AI’s multimodal powers next week!

Q&A Session

- Open floor for questions and clarifications.